

The background of the slide features a light gray field with several thin, curved lines and dashed arcs, creating a sense of motion or data flow. A large, solid green rectangle is centered on the slide, serving as a container for the title and author information.

Data Wrangling: Data Quality

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What is Data Cleaning?

- The most important thing to keep in mind about data cleaning, is that it's an iterative process.
 - We iterate on first detecting, and then correcting bad records.
- Let's look at some examples.
 - We might have text where we expect to find numeric data.
 - So the word two instead of the number 2.
 - Some data items might not be designed according to our specification.
 - They might be missing entire fields.
 - They might have extra fields.
 - Or, for example with JSON documents, fields might not have the structure we expect at all.
 - We might have numbers that are out of range.
 - How many people do you know who are minus three years of age?
 - The mighty outliers compared to a standard distribution.
 - We might have an American expatriate in the London office who did some data entry, and coded in a few dates using the U.S. format when the system was expecting the European format.



Sources of Dirty Data

- There are lots of sources of dirty data.
 - Basically, anytime humans are involved, there's going to be dirty data.
- In some situations, we won't have any data coding standards, or where we do have standards they'll be poorly applied, causing problems in the resulting data.
- We might have to integrate data where different schemas have been used for the same type of item.
- Over time systems evolve. Needs change, and data changes.
- Some of our data won't have the unique identifiers it should.
- Other data will be lost in transformation from one format to another.
- And then of course there's always programmer error.
- And finally, data might have been corrupted in transmission or storage by cosmic rays or other physical phenomenon.

Measuring Data Quality

- Validity: Conforms to a schema
- Accuracy: Conforms to a gold standard
- Completeness: All records?
- Consistency: Matches other data (with sources overlap)
- Uniformity: Some Units

Can you rank how difficult these aspects can be measured from 1 to 5, 1 being the most difficult and 5 the easiest?



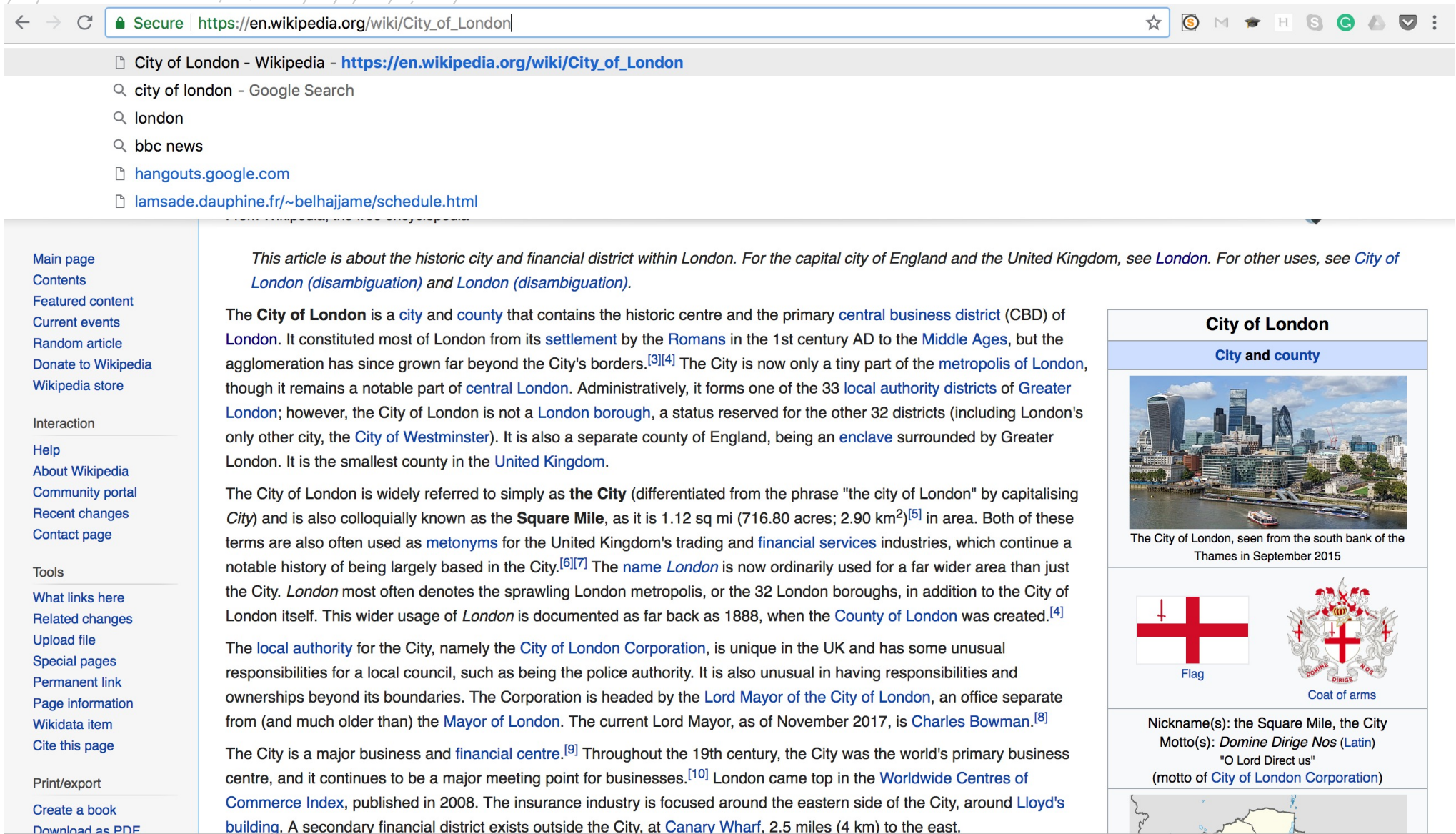
Blueprint for Cleaning

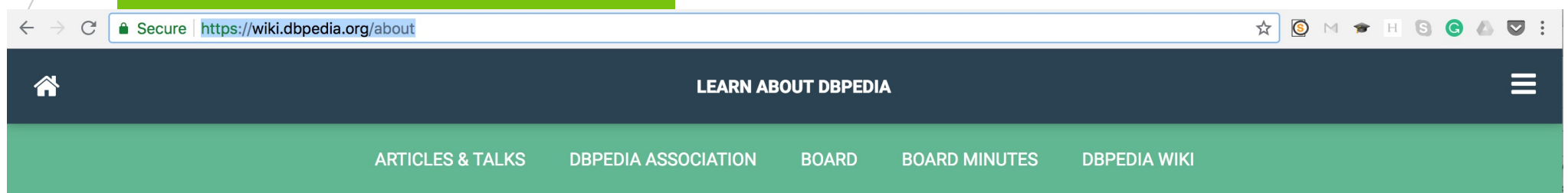
So there's a fairly well established procedure for cleaning data

1. **Audit your data:** In this step what you'll be doing is programmatically checking the data using some validation rules you've written, and you'll create a report on the quality of the data. Now, you might also want to run a statistical analysis in order to, for example, check for outliers.
2. **Create a data cleaning plan:** Having run your audit, then you use that information to create a data cleaning plan. In our data cleaning plan, we want to identify any causes of the dirty data that we're seeing. Now, this is going to be very situation-specific. Then we're going to define a set of operations that will correct our data, again, situation-specific.
3. **Execute the Plan:** Run a few tests to make sure that your data cleaning plan is going to do what you need it to do. Next we're going to execute the plan. Now just to be clear, this is essentially running a script that will execute that set of operations you defined in the data cleaning plan.
4. **Manually Correct:** Now it's possible, in fact it's likely, that we won't have been able to programmatically clean our data entirely. And if it's necessary for you to do so, there maybe a manual correction step, where you go through, or you hire someone to go through, and make any changes that need a human eye in order to correct the data.
5. We may have to iterate more than two iterations until we are confident that the data is ready for analysis.

https://en.wikipedia.org/wiki/City_of_London

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About

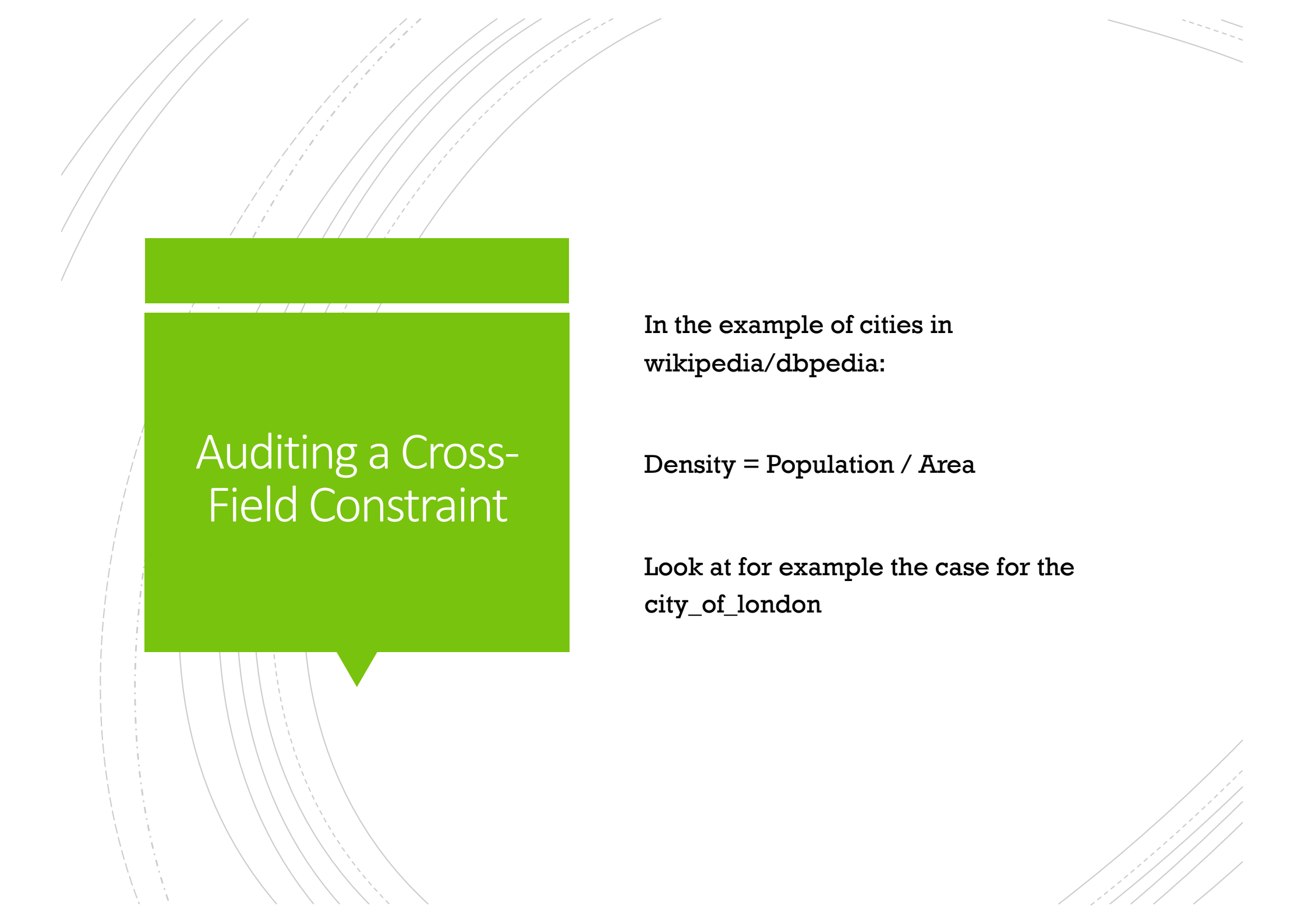
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DBpedia data is served as Linked Data, which is revolutionizing the way applications interact with the Web. One can navigate this Web of facts with standard Web browsers, automated crawlers or pose complex queries with SQL-like query languages (e.g. SPARQL). Have you thought of asking the Web about all cities with low criminality, warm weather and open jobs? That's the kind of query we are talking about.

The DBpedia Knowledge Base

Knowledge bases are playing an increasingly important role in enhancing the intelligence of Web and enterprise search and in supporting information integration. Today, most knowledge bases cover only specific domains, are created by relatively small groups of knowledge engineers, and are very cost intensive to keep up-to-date as domains change. At the same time, Wikipedia has grown into one of the central knowledge sources of mankind, maintained by thousands of contributors.

The DBpedia project leverages this gigantic source of knowledge by extracting structured information from Wikipedia and by making this information accessible on the Web under the terms of the [Creative Commons Attribution-ShareAlike 3.0 License](#) and the [GNU Free Documentation License](#).

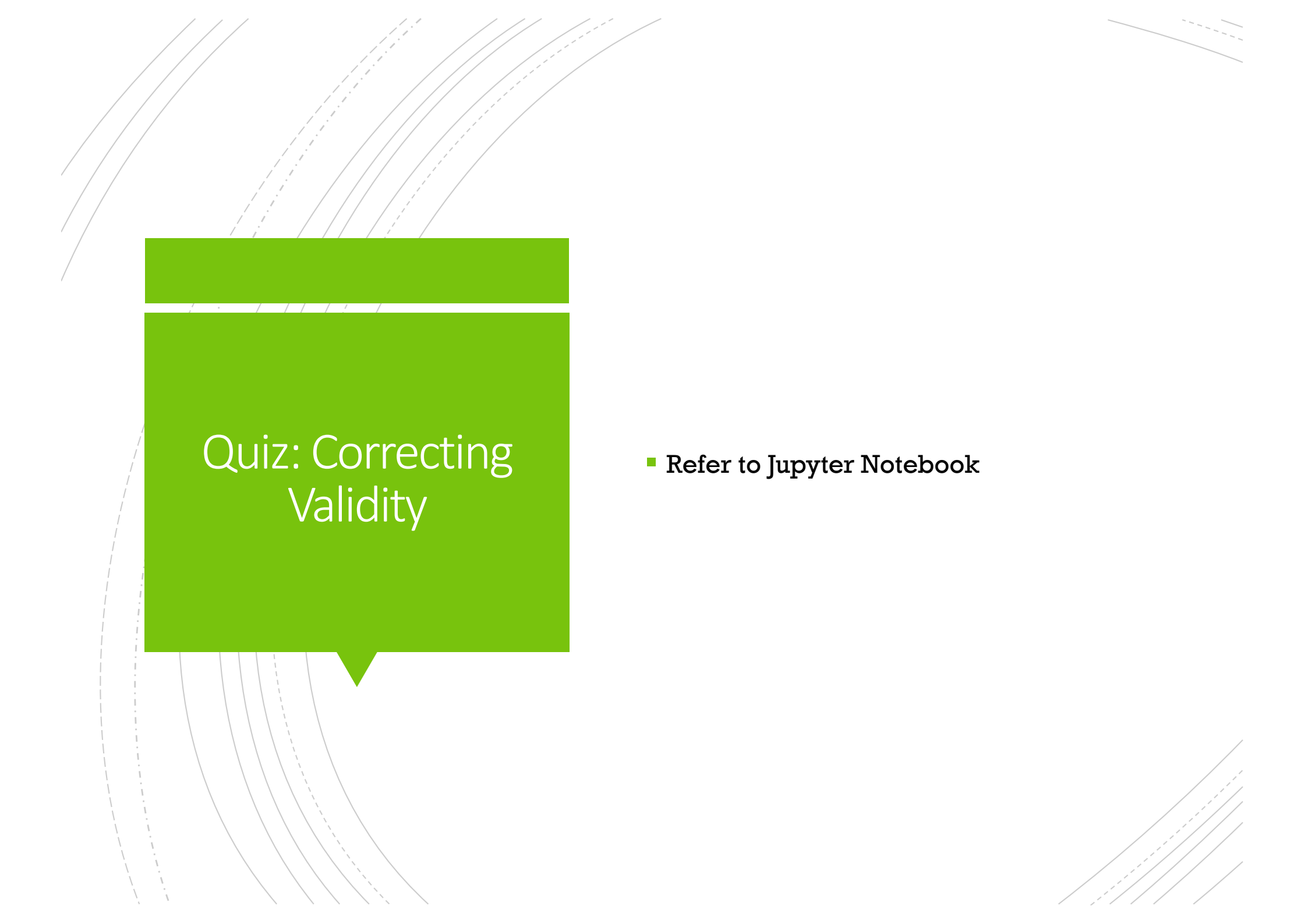
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Auditing a Cross-Field Constraint

In the example of cities in
`wikipedia/dbpedia`:

$\text{Density} = \text{Population} / \text{Area}$

Look at for example the case for the
`city_of_london`

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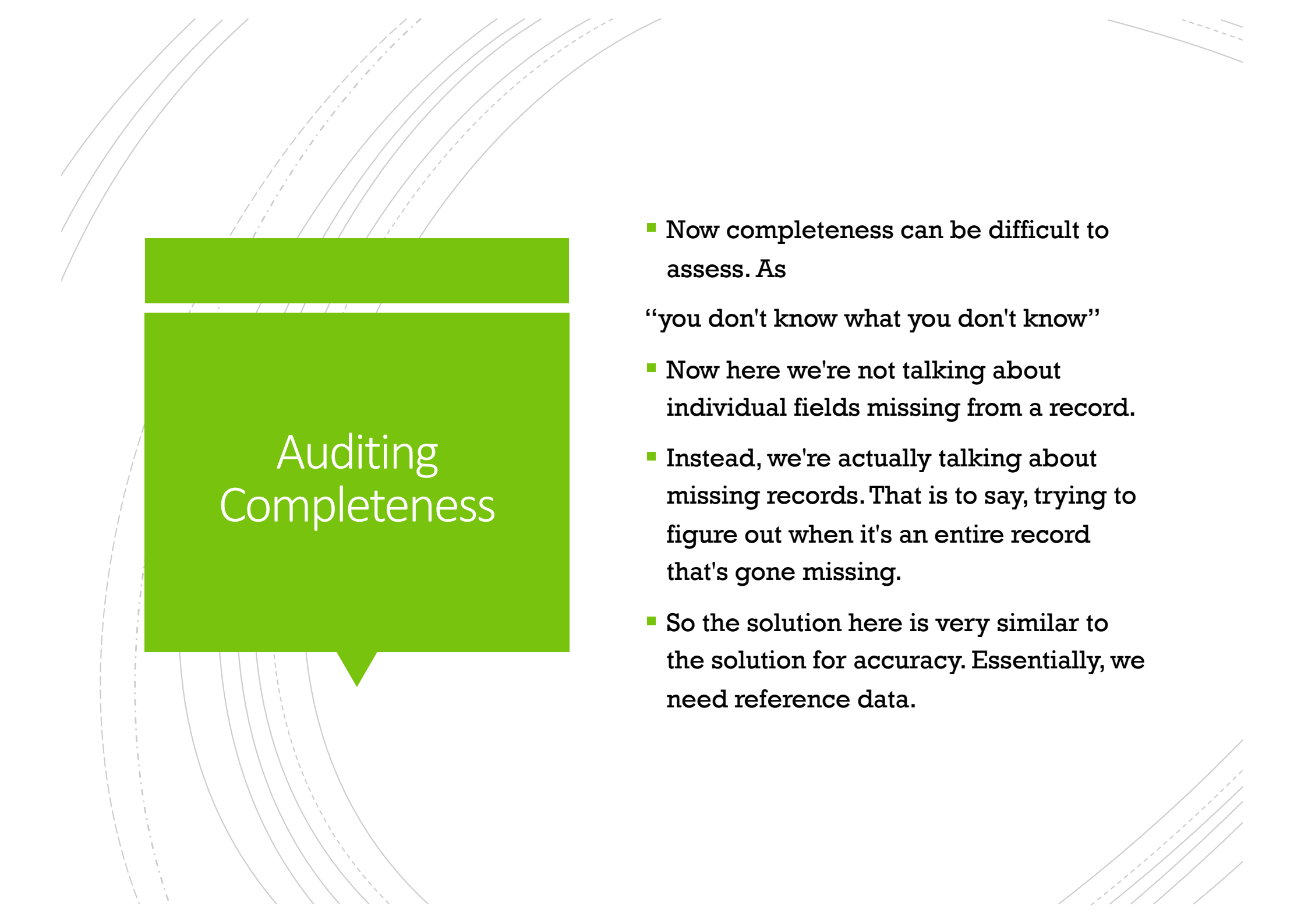
Quiz: Correcting Validity

- **Refer to Jupyter Notebook**



Auditing Accuracy

- Now let's turn our attention to Auditing Accuracy.
- As I mentioned in the overview, this is something that is difficult to do in a lot of situations because it requires gold standard data.
- Let's take a look at our city's info box data set for an example. Returning to the City of London wikipedia page. If we scroll down, we can see that country is one of the fields that's included in the info box data for this, and for most countries in this data set.
- Now it's possible in our data that the countries specified for a city isn't actually a valid country for one reason or another.
- So in order to audit this country field for accuracy, we're going to need some means of programmatically processing it and comparing and the values that we find to some set of data that we can reliably assume is correct.
- The gold standard data that we can use here is some country code data from the ISO. The International Standards Organization.
- This is data that pairs a two digit country code with a country name. So in order to verify these country codes, we're essentially going to do a look up of each name we find in this list of country names we can rely on.

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Auditing Completeness

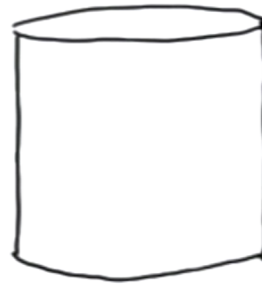
- Now completeness can be difficult to assess. As

“you don't know what you don't know”

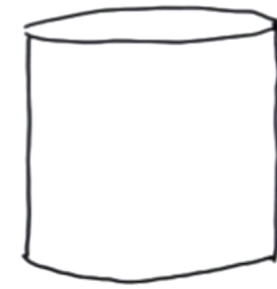
- Now here we're not talking about individual fields missing from a record.
- Instead, we're actually talking about missing records. That is to say, trying to figure out when it's an entire record that's gone missing.
- So the solution here is very similar to the solution for accuracy. Essentially, we need reference data.



COMPLETED
EXAMS



TEST
TAKER
VIDEO



SCREEN
CAPTURE

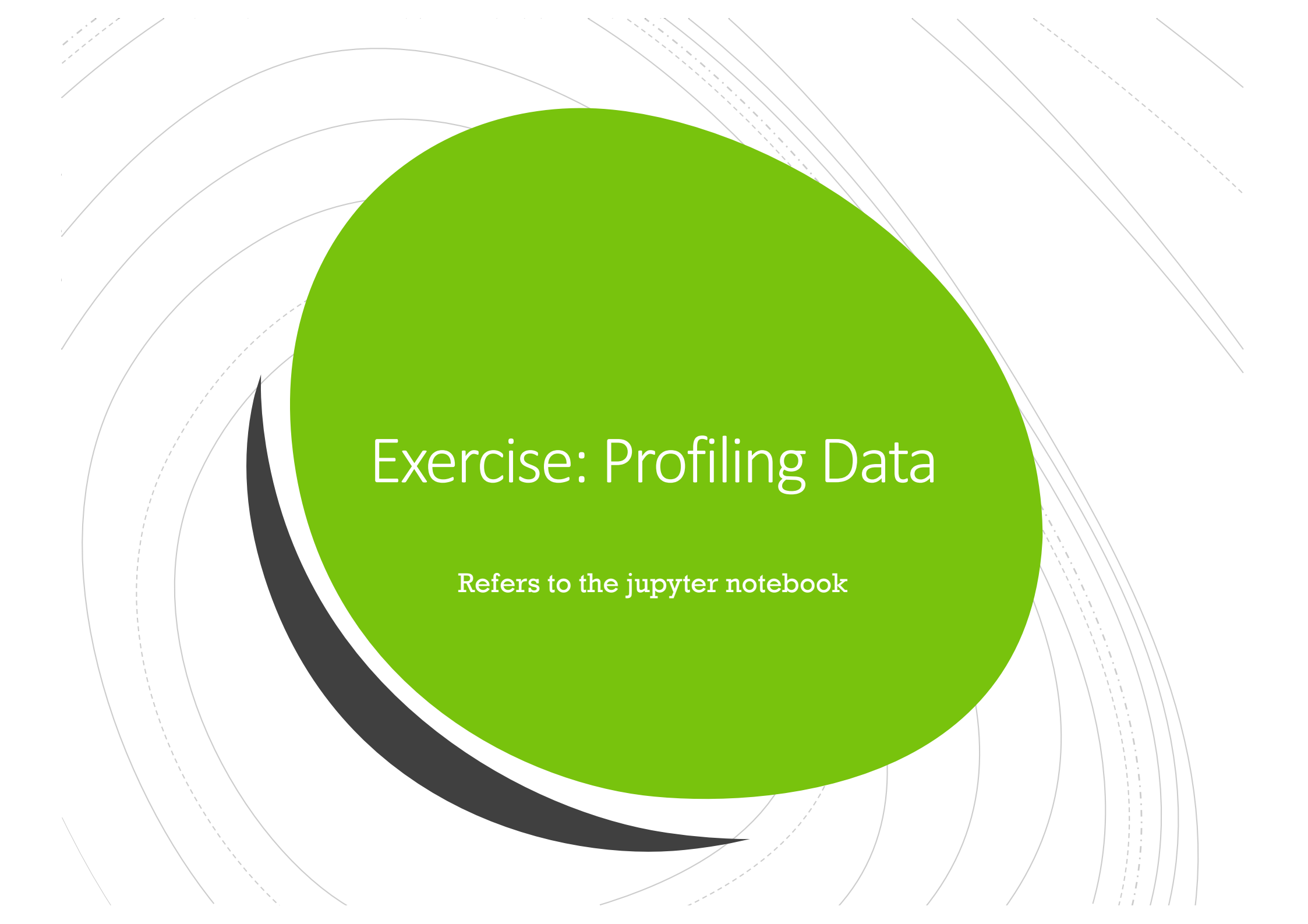
Auditing Completeness: Example

- They must agree on the list of test takers, that is to say, if there is a record for a test taker in any one of these databases, there must be a record for that same test taker in the other two.
- They also need to agree on the duration of the exam session. So the video, here and here, should be approximately the same length and it should agree with the elapsed time that we recorded for a test taker. And, of course, these have to agree approximately.



Auditing Consistency

- Inconsistency occurs when two different entries contradict one another.
- So for example, if a customer is recorded in two different current addresses, we've got a consistency problem.
- Fixing consistency problems really boils down to the problem of who do I trust the most, or more specifically, which data source do I trust the most?
- So with our address example, we might ask ourselves which entry was collected most recently.
- Or if our problem is one that's a little more general say just location information. Another type of question we can ask is which collection method is likely to be most reliable? Because one data source might be populated based on GPS while another is based on user enter data. We might actually have a third that's using IP based location detection and all three of these are contributing location information to our system.



Exercise: Profiling Data

Refers to the jupyter notebook